

Effect of a Disability

ECON 300 – Labour Economics

Winter 2026

General Framework (Both Panels)

- The vertical axis measures **income** (Y).
- The horizontal axis measures **leisure**.
- The downward-sloping straight line represents the **budget constraint**.
- Indifference curves represent preferences over income and leisure.
- Optimal choices occur at points where an indifference curve is tangent to the budget constraint.
- Hours of work are defined as total available time minus leisure.

Panel (a): Reduced Work Time — Partial and Full Disability

Key idea: Disability reduces the individual's *maximum feasible working time*.

- Before disability:
 - The individual can supply up to H_f hours of work.
 - The optimal choice is at point E_o , associated with indifference curve U_o .
 - Hours worked are H_o .
- After disability:
 - The individual's maximum feasible work time is reduced.
 - Under partial disability, maximum hours fall to H_p .
 - Under full disability, feasible work time may fall further.

- The budget constraint becomes **shorter**, ending at H_p instead of H_f .
- Because the old optimum E_o is no longer feasible:
 - The individual chooses a new optimum E_p .
 - The new indifference curve U_p lies below U_o , indicating lower utility.
 - Hours worked fall from H_o to H_p .

Economic interpretation:

- Disability acts as a **constraint on the time endowment**.
- Labour supply falls mechanically because the individual cannot physically work as many hours.
- Preferences and wages are unchanged, but feasible choices are restricted.

Panel (b): Increased Disutility of Labour

Key idea: Disability increases the *disutility of working* without reducing available time.

- The budget constraint remains unchanged:
 - Wage rate is the same.
 - Maximum available time is unchanged.
- Disability increases the physical or mental cost of working:
 - Working becomes more painful or exhausting.
 - Leisure becomes relatively more valuable.
- Preferences change:
 - Indifference curves become **steeper**.
 - Original indifference curve: U_o .
 - New indifference curves after disability: U'_d and U''_d .
- Optimal choice shifts:
 - Tangency moves to the right.
 - Leisure increases.
 - Hours worked fall from H_o to H_d .

Economic interpretation:

- Disability affects **preferences**, not constraints.
- Labour supply falls because the marginal disutility of work rises.
- This is a behavioural response rather than a mechanical restriction.

Comparison of the Two Mechanisms

	Panel (a)	Panel (b)
What changes	Time constraint	Preferences
Budget constraint	Shorter	Unchanged
Indifference curves	Same	Steeper
Reason hours fall	Physical limit	Higher disutility
Nature of response	Mechanical + choice	Choice only

Key Takeaway (Chapter 3)

- Disability can reduce labour supply through two distinct channels:
 1. Reduced capacity to work (time constraint).
 2. Increased disutility of working (preference change).
- Observed reductions in work do not necessarily imply a reduced willingness to work.
- Effective disability and income support policies must account for both mechanisms.